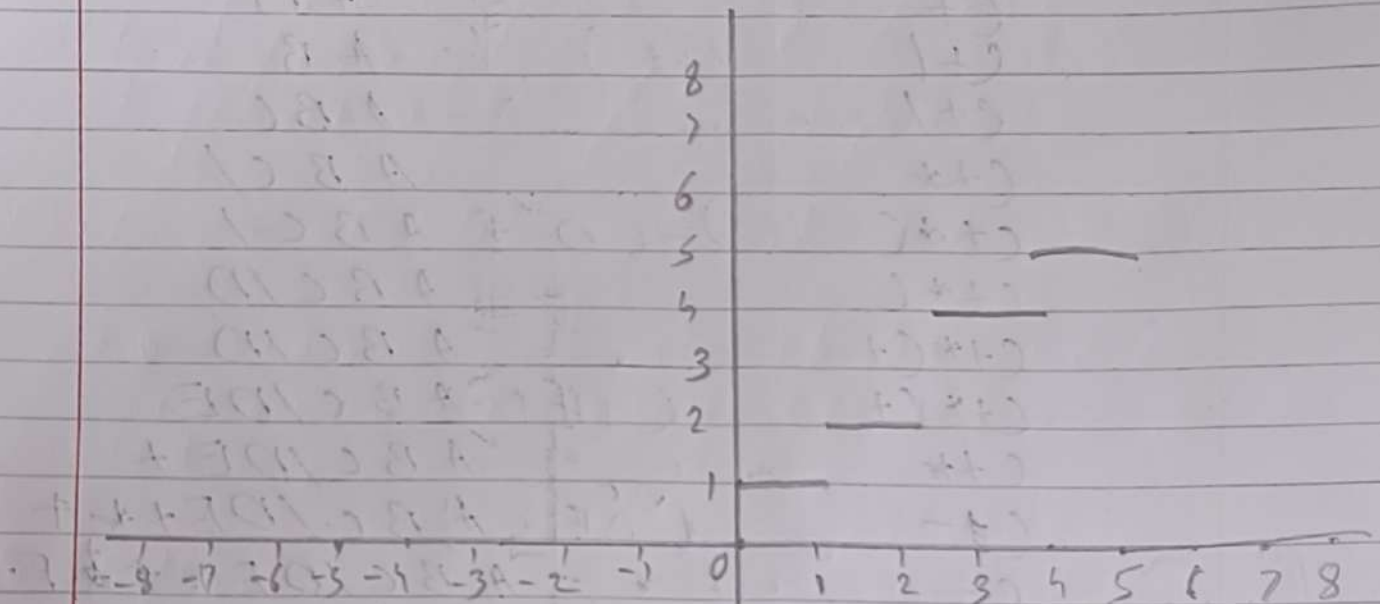


Assignment - 3 202412058

① $f(x) = x^2 + 1$ $g(x) = x + 2$

→ $f \circ g(x) = x^2 + 2x + 5$ and $g \circ f(x) = x^2 + 3$

②



$$\textcircled{3} (1) f(S \cup T) = f(S) \cup f(T)$$

$$\text{let } y \in f(S \cap T) \rightarrow S \cap T \in f(x) = y$$

$$\therefore y \in f(S) \cup f(T)$$

$$\therefore f(S \cap T) \subseteq f(S) \cup f(T)$$

$$\text{Now let } y \in f(S) \text{ or } y \in f(T)$$

$$\therefore y \in f(S) \text{ or } y \in f(T)$$

$$\text{Suppose } y \in f(S) \exists \text{ some } x \in S \ni f(x) = y$$

$$S \subseteq S \cup T, x \in S \cup T, \text{ so } y \in f(S \cup T)$$

$$y \in f(T) \exists \text{ some } x \in T \ni f(x) = y$$

$$y \in f(S \cup T)$$

$$\therefore f(S) \cup f(T) \subseteq f(S \cup T) \dots \textcircled{2}$$

from 1 & 2

$$f(S \cup T) = f(S) \cup f(T)$$

$$(ii) f(S \cap T) \subseteq f(S) \cap f(T)$$

$$\text{Let } y \in f(S \cap T)$$

$\rightarrow \exists$ Some $x \in S \cap T$ s.t. $f(x) = y$
 $x \in S \cap T$ we know that $x \in S$ & $x \in T$

$$\therefore y = f(x) \in f(S) \text{ and } y = f(x) \in f(T)$$

$$\therefore y \in f(S) \cap f(T)$$

$$\therefore f(S \cap T) \subseteq f(S) \cap f(T)$$

(9) Binary Relation on set $A \times B$.

$$\{ \emptyset, \{ (a, a) \}, \{ (a, b) \}, \{ (b, a) \}, \{ (b, b) \}, \\ \{ (a, a), (a, b) \}, \{ (a, a), (b, a) \}, \{ (a, b), (b, a) \}, \{ (a, a), (a, b), (b, a), (b, b) \} \}$$

• Reflexive

$$\rightarrow \{ (a, a), (b, b) \}$$

$$\rightarrow \{ (a, a), (a, b), (b, b) \}$$

$$\rightarrow \{ (a, a), (b, a), (b, b) \}$$

$$\rightarrow \{ (a, a), (a, b), (b, a), (b, b) \}$$

• Symmetric

$$\rightarrow \emptyset$$

$$\rightarrow \{ (a, b), (b, a) \}$$

$$\rightarrow \{ (a, a), (b, b) \}$$

$$\rightarrow \{ (a, a), (a, b), (b, a) \}$$

$$\rightarrow \{ (a, b), (b, a), (b, b) \}$$

$$\rightarrow \{ (a, a), (a, b), (b, a), (b, b) \}$$

• Asymmetric:

- $\{(a, b)\}$
- $\{(b, a)\}$
- $\{(a, a), (a, b)\}$
- $\{(a, a), (b, a)\}$
- $\{(a, a), (b, b), (a, b)\}$
- $\{(a, a), (b, b), (b, a)\}$
- $\{(a, a), (a, b), (b, a), (b, b)\}$

• Anti-Symmetric

- $\emptyset$
- $\{(a, a)\}$
- $\{(b, b)\}$
- $\{(a, a), (b, b)\}$
- $\{(a, a), (b, b), (a, b)\}$
- $\{(a, a), (b, b), (b, a)\}$
- $\{(b, b), (a, b)\}$
- $\{(b, b), (b, a)\}$

(5) $R = \{(1, 2), (2, 1), (2, 3), (3, 5), (4, 1)\}$

Transitive Closure: - $\{(1, 1), (1, 3), (2, 2), (2, 5), (3, 5), (4, 1), (4, 3), (4, 5), (5, 5)\}$

⑥ Smallest Relation $\{(1,2), (1,4), (3,3), (4,1)\}$

④ Reflexive & Transitive

$\rightarrow \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,4), (4,1), (4,2)\}$

⑤ Symmetric & Transitive

④ Symmetric & Transitive

$\rightarrow \{(1,2), (2,1), (1,1), (1,4), (4,1), (4,2), (2,4), (3,3)\}$

⑤ Reflexive, Symmetric & Transitive

$\rightarrow \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,4), (4,1), (4,2), (2,4)\}$

⑦ Find Equivalence Relation $\{0,1,2,3\}$

④ $R_1 = \{(0,0), (0,2), (2,0), (2,2), (2,3), (3,2), (3,3)\}$
Here $(1,1) \notin R_1$

$\therefore R_1$ is not reflexive & Hence not equivalence

④ $R_2 = \{(0,0), (0,2), (2,0), (2,2), (2,3), (3,2), (3,3)\}$

(i) R_2 is reflexive because $(0,0), (1,1), (2,2), (3,3) \in R_2$

(ii) R_2 is symmetric because $(1,3) \in R_2 \& (3,1) \in R_2$
also $(2,3) \in R_2 \& (3,2) \in R_2$

$\therefore$ if $(a,b) \in R \rightarrow (b,a) \in R_2$ is satisfied
for all $(a,b) \in R_2$

$\therefore R_2$ is not equivalence relation